21. TOWER OF HANOI

AIM

To write a Prolog program to implement the Tower of Hanoi problem using recursion.

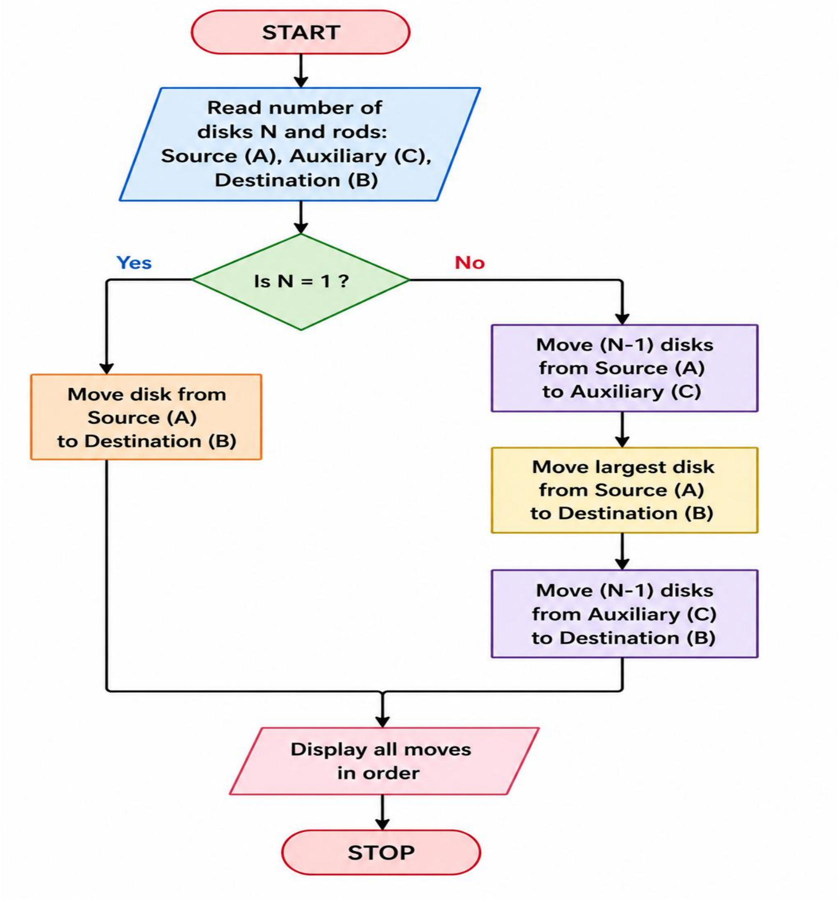
OBJECTIVE

- To understand recursion in Prolog.
- To solve the Tower of Hanoi problem using three rods.
- To generate the sequence of moves required to transfer all disks.
- To follow the rule that a larger disk cannot be placed on a smaller disk.

ALGORITHM

1. Start.
2. Read the number of disks N.
3. If N = 1, move the disk from source to destination.
4. Otherwise, move N-1 disks from source to auxiliary rod.
5. Move the largest disk from source to destination.
6. Move the N-1 disks from auxiliary to destination.
7. Repeat recursively until all disks are moved.
8. Display all moves.
9. Stop.

FLOWCHART:



CODE:

```
hanoi(1, A, B, _) :-
    write('Move disk from '), write(A),
    write(' to '), write(B), nl.
hanoi(N, A, B, C) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, A, C, B),
    hanoi(1, A, B, C),
    hanoi(N1, C, B, A).
```

OUTPUT:

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File  Edit  Settings  Run  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/tower of hanoi.pl compiled 0.00 sec, 2 clauses
?-  hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true
```

RESULT:

Thus, the Tower of Hanoi problem was successfully implemented in Prolog using recursion, and the sequence of moves required to transfer the disks was obtained.